Experiment 15: H_2O_2 + Boric Acid — SVEs Surviving Cyclic Tidal Exposure

"Tidal Cycles in a Plastic Bottle: SVEs That Refuse to Dissolve"

Experimental Protocol

Container: Standard plastic bottle (garrafa plástica). **Liquid:** 50/50 mixture of H_2O_2 3% (antiseptic hydrogen peroxide) and boric acid solution 3% (água boricada, antiseptic). **Heat source:** None — ambient conditions only. **Recording:** iPhone slow-motion video, frame-by-frame analysis. **Analysis:** Single complete tidal cycle tracked across 9 frames (1 collage). SVE positions annotated with red circles; liquid level annotated with yellow line. Multiple complete cycles visible in the full video; only the highest-amplitude SVE (the most superior one) eventually detaches after many cycles — the other 3 survive the entire recording. **Video:** https://youtube.com/shorts/aXFAot2Q-QE

Complete Data: Experiment 15 (1 Collage, 9 Frames)

Coll.	Fr	Frame #	Time	Phase	Observation
C1	F1	93	03:089	OVERVIEW	General view from distance. Plastic bottle with H_2O_2/boric acid mixture. Multiple SVEs visible adhered to the interior plastic wall, submerged in liquid. Cluster of 4 SVEs in focus. Liquid level stable.
C1	F2	262	08:700	HIGH TIDE	Close-up. All 4 SVEs fully submerged inside the liquid. Yellow line marks liquid level at its highest position — maximum submersion. SVEs adhered to plastic wall, spherical, intact.
C1	F3	263	08:733	RECEDING	Liquid level (yellow line) descending. Tidal recession in progress. SVEs still submerged but liquid retreating from them. Movement is FAST — single frame captures significant level change.
C1	F4	264	08:767	PARTIAL	The most superior SVE is now ABOVE the liquid level — exposed to the air inside the bottle. The lower 3 remain at the liquid level. First liquid-air interface crossing for the top SVE. Classical prediction: exposed bubble should rupture/evaporate.
C1	F5	265	08:800	EXPOSED	All 4 SVEs now ABOVE the liquid level — fully exposed to the air phase inside the bottle. Yellow line shows liquid level well below the cluster. All 4 maintain spherical integrity. No rupture, no dissolution, no evaporation.
C1	F6	266	08:833	LOW TIDE	Maximum recession — liquid level at its lowest point. All 4 SVEs at maximum distance from the liquid surface. Maximum air exposure duration. All intact. This is the mirror of F2: same entities, opposite phase.
C1	F7	267	08:867	RISING	Liquid level rising — tide returning. SVEs at the liquid level again. Second interface crossing: air → liquid. Entities being re-submerged. Structural integrity maintained through re-wetting.
C1	F8	268	08:900	SUBMERGED	All 4 SVEs fully submerged again. Liquid has risen back above the cluster. Complete cycle: submerged → exposed → submerged. All entities intact — zero structural degradation.
C1	F9	269	08:933	DEEP	Liquid level at its highest point of returning tide — SVEs at maximum depth. One complete tidal cycle documented. In the full video, this cycle repeats many times. Only the most superior SVE eventually detaches; the other 3 survive all cycles until recording ends.

Photographic Documentation

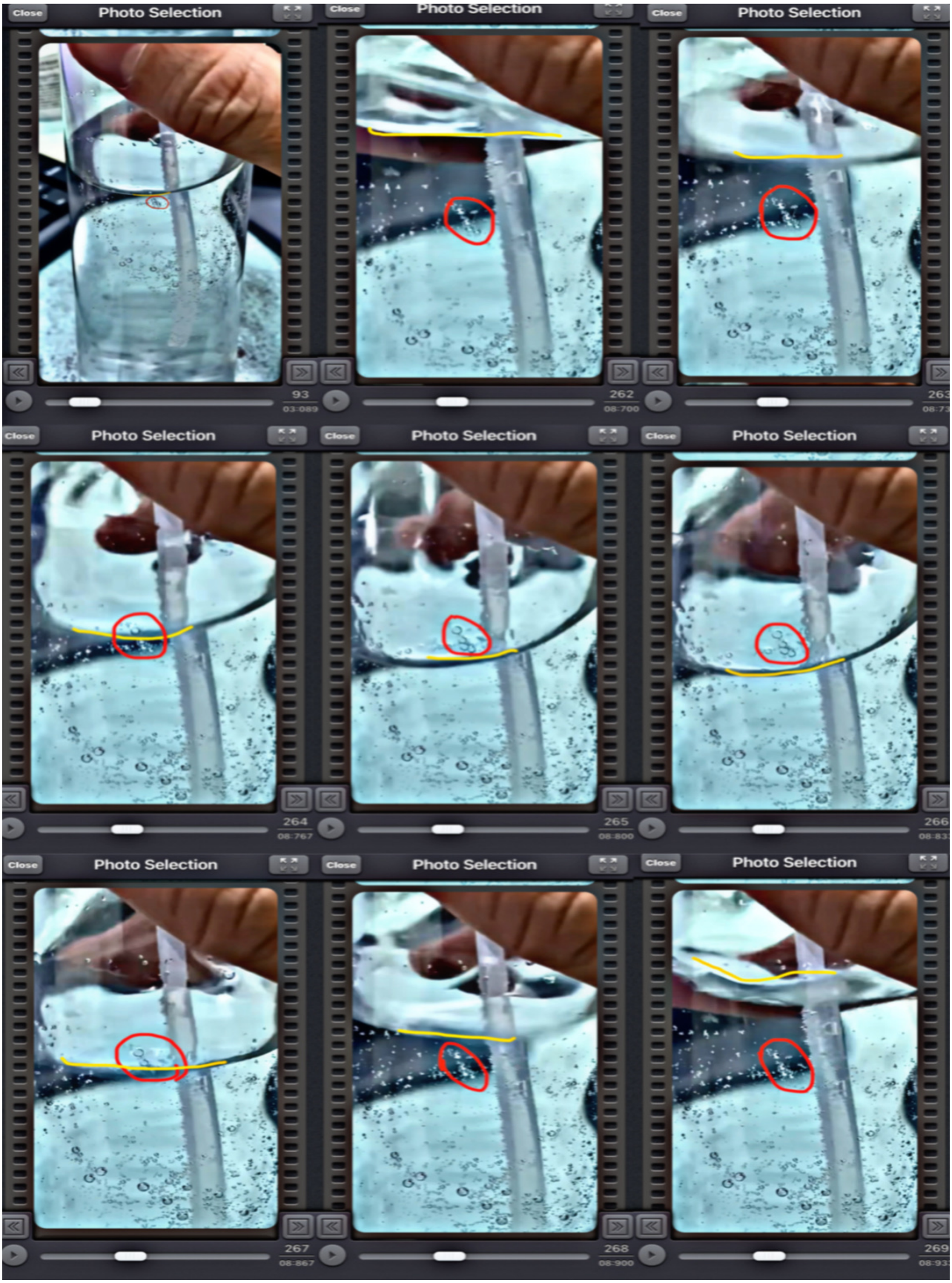

Figure E15-C1 — Complete tidal cycle (9 frames). Red circles: 4 SVEs adhered to the interior plastic wall of the bottle. **Yellow line:** liquid level. F1 (frame 93): overview from distance. F2 (262): high tide — all SVEs submerged. F3 (263): liquid receding. F4 (264): top SVE crosses interface — exposed to air. F5 (265): all 4 SVEs above liquid level — fully exposed to air inside the bottle. F6 (266): low tide — maximum air exposure, all intact. F7 (267): tide rising — second interface crossing. F8 (268): all SVEs re-submerged. F9 (269): deep — cycle complete. The cycle repeats many times in the full video. Only the most superior SVE eventually detaches after numerous cycles; the remaining 3 survive every cycle until the recording ends. Each cycle involves two complete liquid-air interface crossings per SVE — rapid, violent, and repeated.

Anomaly Analysis — Experiment 15

Anomaly E1: Repeated liquid-air interface crossings without rupture.

Each tidal cycle subjects the SVEs to two interface crossings: liquid→air (exposure) and air→liquid (re-submersion). For a gas bubble adhered to a surface, crossing the liquid-air interface means the liquid film maintaining the bubble's structure is removed. Without the liquid film, the gas pocket has no confining mechanism — it disperses. The observed SVEs cross this interface repeatedly — dozens of times across the full recording — without any structural degradation. This is consistent only with entities whose cohesion is mechanical/structural, not film-dependent. The interface crossing that should destroy them has no effect.

Anomaly E2: Survival of rapid, violent tidal oscillation.

The tidal oscillation is not gentle. Frame-by-frame analysis shows the liquid level changing significantly between consecutive frames (33 ms intervals in slow motion). The movement is described as BRUSCO/VIOLENTO/RÁPIDO — abrupt, violent, fast. This subjects the SVEs to repeated hydrodynamic shear forces at the interface — precisely the conditions that maximize the probability of bubble disruption. Classical bubble dynamics predicts that oscillatory shear at a gas-liquid interface accelerates film thinning and rupture. The SVEs not only survive this shear — they survive it cyclically, many times, with only the most exposed entity eventually detaching (not rupturing — detaching) after numerous cycles.

Anomaly E3: Structural integrity maintained through air exposure.

During the LOW TIDE phase (F5-F6), all 4 SVEs are fully above the liquid level, surrounded by air inside the sealed bottle. For a gas bubble, removal from the liquid phase eliminates the only structural element (the liquid film). A gas pocket without its confining film is just... gas. It disperses instantly. The observed SVEs maintain their spherical geometry, their size, their adhesion to the plastic wall, and their relative positions throughout the entire air-exposure phase. They then re-enter the liquid phase with zero observable change. This is the behavior of a solid or semi-solid structure, not a gas pocket.

Anomaly E4: Adhesion to plastic substrate across phase changes.

The SVEs remain adhered to the interior plastic wall of the bottle throughout the entire recording — through multiple complete tidal cycles, through repeated liquid-air-liquid transitions, through violent hydrodynamic shear. Classical bubble adhesion to surfaces is mediated by the liquid film: the contact angle between gas, liquid, and solid determines adhesion force. When the liquid is removed (air exposure), the adhesion mechanism vanishes. The SVEs maintain adhesion in BOTH phases — submerged in liquid AND exposed to air. This requires a surface interaction independent of the liquid film — consistent with mechanical/chemical adhesion of a solid or semi-solid entity, not gas-film-mediated contact.

Anomaly E5: Differential survival — structural hierarchy.

After many complete tidal cycles, only the most superior SVE (the one experiencing the greatest amplitude of exposure) eventually detaches. The 3 lower SVEs survive every cycle until the recording ends. This differential survival implies: (a) The entities have varying degrees of structural robustness — a property of solid/semi-solid structures, not gas pockets (all gas bubbles have identical mechanical properties at a given pressure). (b) The detachment mechanism is mechanical fatigue, not film rupture — the entity weakens progressively over many cycles before failing. Gas bubbles do not exhibit fatigue; they either exist or they don't. The observed gradual degradation is a hallmark of solid-state mechanics.

Systematic Falsification — Experiment 15

Classical Prediction	Observation	Status
Gas bubbles rupture when the liquid film is removed (air exposure)	SVEs survive complete air exposure, maintain spherical form, re-submerge intact	**FALSIFIED**

Classical Prediction	Observation	Status
Oscillatory shear accelerates bubble rupture at the interface	SVEs survive dozens of rapid, violent tidal cycles — repeated shear has no effect	**FALSIFIED**
Bubble adhesion to surfaces is mediated by the liquid film (contact angle)	SVEs remain adhered to plastic in BOTH liquid and air phases — film-independent adhesion	**FALSIFIED**
All gas bubbles at a given pressure have identical mechanical properties	Differential survival: top SVE fails after many cycles; 3 lower SVEs survive indefinitely	**FALSIFIED**
Gas bubbles do not exhibit mechanical fatigue	Top SVE detaches after progressive weakening over many cycles — solid-state fatigue behavior	**FALSIFIED**

Cross-Experiment Consistency

Pattern	Previous Experiments	Experiment 15
SVEs survive interface crossings	Exp. 14: SVE breaches surface, arcs through air, dives back — 3 interface crossings in one trajectory.	SVEs survive dozens of cyclic liquid↔air crossings — repeated, violent, sustained.
Structural integrity beyond gas-phase predictions	Exp. 9: microscopic SVEs in saliva show cohesion, motility. Exp. 14: ballistic arcs, sinking.	SVEs maintain form through air exposure where no liquid film exists to confine them.
Spontaneous formation without thermal gradient	Exp. 14: SVEs in carbonated Pepsi at ambient temperature, no heating.	SVEs in H_2O_2/boric acid at ambient temperature — second non-thermal system.
Universality across chemical systems	Water, butter, legumes, herbs, egg whites, Pepsi Black (CO_2).	H_2O_2 + boric acid — eighth distinct chemical system exhibiting SVE phenomena.

Experiment 15 introduces the most stringent test of SVE structural integrity to date: cyclic, rapid, violent exposure to repeated liquid-air interface crossings. Where Experiment 14 documented a single dramatic trajectory (one entity, one arc, one dive), Experiment 15 documents sustained endurance — the same entities surviving dozens of complete tidal cycles, each involving two interface crossings and significant hydrodynamic shear. The differential survival (1 of 4 entities eventually failing after many cycles while 3 survive indefinitely) introduces mechanical fatigue as an observable — a property exclusive to solid-state materials and entirely absent from gas-phase physics. The system requires no heating, no exotic chemicals — just two common antiseptic solutions mixed in a plastic bottle. The SVEs form spontaneously, adhere to the plastic substrate, and demonstrate structural permanence across repeated phase-boundary violations. This is the eighth distinct chemical system in the Kitchen Thermodynamics series to exhibit SVE-mediated anomalies, and the second (after Exp. 14) to do so without any applied thermal gradient.

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Video: https://youtube.com/shorts/aXFAot2Q-QE
Supplementary materials: Google Drive · YouTube: @5incere5cience · TikTok: @5incere5cience

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